

Statement of Work
Chemistry Analyzers
General Leonard Wood Community Hospital
Fort Leonard Wood, Missouri

EQUIPMENT SALIENT CHARACTERISTICS

Require two integrated immunochemistry and clinical chemistry analyzers to work in conjunction with each other in a single unit which must fit within a workspace of no more than 22 feet. This workspace also incorporates the cooling requirement of one (1) foot clearance around the unit for safety and cooling purposes. The Contractor shall provide Chemistry /Immunoassay Reagents (including equipment, maintenance, and support at no extra cost to the Government) in support of the General Leonard Wood Community Hospital, Fort Leonard Wood, MO 65473. The Contractor shall be responsible for furnishing maintenance, services, repairs, and parts as required to maintain the systems in optimal operating conditions.

The Contractor shall provide system redundancy (to ensure continuous operations), training, consumables, supplies, reagents, unscheduled maintenance, preventive maintenance for the equipment and 24-hour telephonic assistance. All equipment displays and supporting literature must be in English language. Contractor shall include delivery, installation, all calibrators/standards, reagents/reagent packs, and quality control materials necessary to perform the specified assays, initial Method Validation per CLIA/CLSI and QM/Pathology guidelines and analyzer removal at the termination of the contract. All costs will be calculated on a cost per patient reportable or patient billable basis based on the number of patients listed for each Mandatory Test or Optional Test listed in Attachment 1. Costs shall be amortized over the total number of tests described in this PWS and factored into the cost per reportable or patient billable.

During the installation and validation of the equipment and tests to be performed, the Contractor will schedule meetings with technical experts of the Laboratory to discuss milestones and to provide updates on completion date. Following the completion of the installation of the new equipment and validation of the tests to be performed, the Contractor will meet with technical experts from the Laboratory at times mutually agreeable to discuss areas of interest to the Laboratory or hospital staff.

Method Validation will be conducted entirely by the Contractor in accordance with method validation protocols established by the Department of Pathology's Quality Management Program (these protocols are in accordance with applicable standards as described by the Clinical Laboratory Standards Institute (CLSI) and/or required by the College of American Pathologists (CAP).

Method Validation will include but not be limited to, reference range verification, demonstration of linearity with limit of quantitation verification (LOQ), limit of detection verification (LOD), reportable range, correlation with current methods (comparison of methods study), validation of analytical measurement range (AMR), validation of clinical reportable range (CRR), validation of imprecision (within day and between day) and accuracy (recovery study), plasma/serum correlation studies, and establishment of initial control ranges using control materials, to include third party controls, as designated by the Laboratory Director.

The Method Validation will be completed and approved by the Laboratory Director prior to reporting of patient results. Payment will not begin until the analyzer system has been approved for reporting of patient results.

The Contractor shall provide a qualified Technical Service Representative, who possesses a thorough knowledge of method validation and is familiar with the use of EP Evaluator® as a statistical evaluation software to provide technical assistance for the method validation studies. The assigned Technical Service Representative will be responsible for the entire Method Validation.

All reagents, calibrators and linearity materials used in the Method Validation Study will be provided by the Contractor at no additional cost to the Government, to include third party reagents and controls as indicated by the Government and agreed upon by the Contractor. These materials will not be used in the analysis of specimens for patient care.

When new tests or new Generations of reagents are available/added, the Contractor will perform all necessary validation studies and supply all necessary reagents, calibrators, linearity materials.

The Government's Technical Supervisor will obtain patient specimens for use in the Method Validation. In the event specimen requirements for correlation studies cannot be met (e.g., insufficient number of patients to challenge the extremes of a curve), the Contractor shall assist in procuring additional specimens. These specimens may be obtained from other laboratories or from Contractor sources.

The Contractor shall provide a qualified Technical Service Representative to perform bi-annual instrument to instrument correlations, AMR/linearity correlations and carry-over studies.

Instrumentation:

Automated units must be dry systems ONLY with minimal reagent storage requirements. Analyzer(s) does not require plumbing such as: a deionized water system, drains, stirring mechanisms, water baths, re-used & washed cuvettes, mixing assemblies, and/or floor vents.

Fully automated platform that must perform chemistry and immunochemistry test on a single instrument and must include the following:

Individual analyzers that are fully automated, have discrete random-access capabilities for integrated chemistry, immunometric, potentiometric, and colorimetric assays within the same instrument, i.e. not separate chemistry, immunometric, potentiometric, and colorimetric analyzers.

Must be random-access, discrete analyzer system with STAT interrupt capability and a primary tube sampling system capable of handling 100mm x 16mm, 100mm x 13mm, 75mm x 13mm, 64mm x 10.25 and 46mm x 10mm (microtainer) test tubes with standard or Hemoguard caps and screw caps. The analyzer will be capable of accommodating both Becton-Dickinson and Greiner evacuated tubes. The analyzer(s) must utilize tube sample metering with clot and bubble detection.

Priority run interruptions, STAT 30-minute turnaround time, performing linearity, precision and verification studies, over range auto dilutions, re-runs, reflex testing and panic repeats without operator intervention, and pediatric (micro samples) low volume capabilities.

Analyzer minimum actual through-put: Testing throughput for each Integrated Analyzer must exceed 1,200 tests per hour and the turnaround time for a Chem 8 panel (BMP) must be less than or equal 10 minutes.

Analyzer must be capable of Chemistry and Immunochemistry tests from same sample tube without technician intervention.

The analyzer has zero sample and reagent carryover by utilizing disposable tips. The system must not use reusable probes to aspirate the sample or reagent. The analyzer must not require carryover studies.

Initial startup from the shutdown state must not exceed 20 minutes.

Daily maintenance for each Integrated Analyzer must not exceed 10 minutes of operator time and 15 minutes of automated time.

Weekly maintenance for each Integrated Analyzer must not exceed 20 minutes of operator time and 30 minutes of automated time.

Un-interrupted power supply (UPS) capable of providing all necessary electrical power to all components of the chemistry/immunoassay analyzer to include data management system, for 15 to 20 minutes. The Contractor is responsible to ensure the UPS remains operable throughout the length of the contract.

The Contractor shall provide two (2) paper and two (2) electronic (CD) copies of the Operator's and Maintenance manuals and a statement of the warranty terms in the English language. The level of detail required of appropriate instrument maintenance manuals extends only to routine maintenance procedures performed by the user. All instrumentation specified is non-Government owned leased equipment that is serviced and maintained by the Contractor. Printer to produce hard copy patient reports and other reports generated via the analyzer system's data management system.

Minimum capacity of 80 on board specimen samples.

Each analyzer must not have a BTU output of more than 12,500.

Capability to add/change out reagents/reagent packs with only a minimal calibration/quality control requirement.

Must have the on-board ability to maintain reagents at the proper storage temperatures for the assays listed in Attachment 1.

Minimum 150 reagent positions.

Operating System:

The analyzer operating system shall include the capability to track and store, at a minimum 90 days of calibration, quality control (QC) and patient result data and provide for visual display of graphs and curves for QC data.

The analyzer operating system shall have the capability to incorporate Bio-Rad's on-line/on-board QC analysis package, Unity™ Interlaboratory Program and Unity QC Data Management Solutions.

The Contractor shall provide Unity™ Interlaboratory Program and Unity QC Data Management Solutions- Unity Real Time and Bio-Rad QC material for all assays.

Must have the ability to read/write quality control, calibrations, and patient results to DVR or USB drive for peer group correlation/review purposes.

Operating system shall interface with the customer's laboratory information system (LIS) via RS 232C bidirectional interface and accommodate American Standard Code for Information

Interchange (ASCII) query capability to process specimen identification and work orders from the LIS (Laboratory Information System).

Must be capable of maintaining calibration curves for all assays listed in Table 2 simultaneously.

Data management system to control the operation of the analyzer.

Data management system must be windows-based. Must be approved and tested with MHS Genesis.

On-instrument shelf-life of the reagents/reagent packs capability.

Ability for auto reflex dilutions.

Ability for automatic reflex testing.

Programming capability to set QC rules or contractor provided middleware that would satisfy the requirement.

STAT panel includes the assays listed in Table 1. These assays will be analyzed from the same specimen with a maximum turn-around-time (pre-analytical processing to certification) of 30 minutes.

Table 1. – Assays that must be completed from the same specimen within 30 minutes as a STAT panel.	
Alanine Amino Transferase (ALT)	Creatinine Kinase, Total (CK)
Aspartate Amino Transferase (AST)	Creatinine Kinase, MB Fraction (CK-MB)
Bilirubin (Direct/Conjugated)	Glucose
Bilirubin (Total)	Human Chorionic Gonadotropin (hCG)
Brain Natriuretic Peptide (BNP) or NT-pro BNP	Potassium
Calcium	Protein, Total
Carbon Dioxide (CO2)	Sodium
Chloride	Troponin I or T
Creatinine, enzymatic	Urea Nitrogen
Phosphorous	Magnesium
Amylase	Lipase
Ammonia	

The QC software shall provide the capability to write user defined Westgard QC Rules.

QC Software shall integrate with the result manager for auto-verification of results, allow viewing and maintaining of QC data across multiple levels and lots for connected analyzers, provide a mechanism to maintain an action log of comments from QC review, provide real-time access to QC charts, graphs (Levy-Jennings) and provide the ability to add comments, and eliminate data point(s) from statistics.

QC software will include the ability to apply Westgard quality control result decision rules in real-time, with the ability of the user to individually select the rules to be applied.

Store patient results for a minimum of 12-months to facilitate auto-verification/auto-result.

Must be automation-ready to allow for future connection of various laboratory instruments, pre and post analytic instruments (i.e. sample manager and /or line system).

Analyzer must be able to determine hemolysis, icterus and turbidity (HIT) levels without additional patient sample, reagents, and consumables with no effect on throughput. The analyzer must flag only the assay(s) results that may be affected by the interferent for both chemistry and immunoassay testing. The HIT flags and level must have the ability to transfer to the LIS or data management system.

Analyzer must be able to connect via internet connection or DSL via VPN secured connection between Contractor and site to provide technical support, data exchange, and software updates.

REAGENT SALIENT CHARACTERISTICS

All assays must be FDA approved.

Assays listed in Attachment 1 must be present on the analyzer at all times.

On board reagent stability must be at least, but no less than, seven days.

Assays listed in Table 2.1 will be approved by the FDA for serum/plasma samples.

Assays listed in Table 2.2 will be approved by the FDA for urine samples.

Assays listed in Table 2.3 will be approved by the FDA for CSF samples.

Assays listed in Table 2.4 –will be approved by the FDA for Whole Blood samples.

Table 2.1 –Assays That Must be Approved for Use With Serum/Plasma Samples		
ACETAMINOPHEN	HDL- HIGH DENSITY LIPOPROTEIN	VANCOMYCIN
ALANINE AMINOTRANSFERASE	Iron	B12/FOLATE
ALBUMIN	LACTATE	BETA HCG
ALKALINE PHOSPHATASE	LACTATE DEHYDRONASE	CORTISOL
AMMONIA	LDL Cholesterol	CREATINE KINASE-MB
AMYLASE	LIPASE	Estrogen -E2
ASPARTATE AMINOTRANSFERASE	LITHIUM	FERRITIN
BILIRUBIN TOTAL	MAGNESIUM	FOLLICLE-STIMULATING HORMONE - FSH
CALCIUM	PHENYTOIN	FREE TRIIODOTHYRONINE - FT3
CARBON DIOXIDE	PHOSPHORUS	LUTEINIZING HORMONE- LH

CHOLESTEROL	POTASSIUM	ProBNP
CK -Creatine Kinase	SALICYLATE	PROCALCITONIN- PCT
CLORIDE	SODIUM	PROGESTERONE
C-REACTIVE PROTEIN	THEOPHYLLINE	PROLACTIN
Creatinine	TIBC/UIBC	PROSTATE SPECIFIC ANTIGEN- PSA
DIGOXIN	TOTAL PROTEIN	TESTOSTERONE
DIRECT BILIRUBIN	TRIGLYCERIDES	THYROID STIMULATING HORMONE -TSH
ETHANOL/ALCOHOL	UREA	THYROXINE- FT4
GAMMA GLUTAMYL TRANSFERASE	URIC ACID	TRIIODOTHYRONINE- T3
GENTAMICIN	VALPROIC ACID	TROPONIN I/ TROPONIN T
GLUCOSE		

Table 2.2 – Listing of Assays That Must be Approved for Use With Urine Samples

Amylase	Phosphorus
Calcium	Potassium
Chloride	Sodium
Creatinine	Total Protein
Glucose	Urea
Magnesium	Uric Acid
Microalbumin	

Table 2.3 – Listing of Assays That Must be Approved for Use With CSF

Glucose	Protein
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Table 2.4 – Listing of Assays That Must be Approved for Use With Whole Blood

A1C

Reagents and consumables must have a minimum shelf life upon receipt by the Government of at least 60 days.

Reagents provided will be identical to those used in the FDA 510K approval application.

The reagents/reagent packs will be provided as follows:

Be provided by the manufacturer of the equipment. May not be a “user defined” assay.

Be marked with the required storage temperature.

Must have a bar-code identification containing the lot number, expiration date, and unique identification code.

Reagents should be “ready-to-use” and require no user performed reagent preparation for any standard testing method.

Reagent kits must contain all calibrators necessary to calibrate each analyte lot.

Delivery of reagents will not exceed 7 calendar days from the date of request.

Reagents and consumables must have the ability to be added on-the-fly with no interruption to patient sample testing allowing for efficient and continuous testing during peak testing periods.

Provide all consumables other than reagents as no charge items.

When the Government performs troubleshooting with the Contractor, the Contractor shall replace all reagents, calibrators, controls and any other materials used at no additional cost to the Government.

MAINTENANCE

At the start of the contract, the Contractor shall have a contract representative and alternate responsible for scheduling and coordination of service calls. The name and phone number of the designated individual and alternate shall be provided to the Chief, Medical Maintenance Branch.

Contractor shall deliver all new equipment to the Property Management Section at the beginning of the contract. Clinical Engineering Section will perform acceptance inspection and establish maintenance records for equipment. These records will be maintained for the duration of the contract period. Clinical Engineering will maintain historical data that shows the acceptance inspection and services performed on the equipment by Medical Maintenance and/or the MFG as long as the equipment is in the Government Facility.

Any and all equipment contracts to include, lease, cost per reportable, loan, etc., must be reviewed and approved by the Chief, Clinical Engineering for purposes of maintenance and accountability, whether accomplished by in-house maintenance or contract personnel.

The GLWACH Medical Maintenance Branch will first evaluate any equipment malfunction and facilitate repairs (screening process). If a service call is necessary, the Contractor shall provide unlimited service calls.

The Contractor shall provide direct customer service and support, 24 hours/7 days a week, to include

hotline telephone service and on-call emergency repair service support, to assist operators in n correcting equipment operation problems.

Preventive maintenance services shall be performed in accordance with the manufacturer's standards/ procedures. A preventive maintenance service shall include, but is not limited to, safety, calibration, complete operational testing, lubrication, adjustments and cleaning of equipment to which the operator does not have access. This also includes the installation of all non-operator parts required to ensure proper operation.

The contractor shall notify the Point of Contact of the exact date and time for performance of a preventive maintenance service at least two (2) days prior to the scheduled preventive maintenance services.

The contractor's qualified service technician shall be onsite to provide routine services, service requiring immediate repair and one (1), two (2) or four (4) scheduled services for preventive maintenance (as specified in the Contractor's instrument maintenance manual), safety and calibration as required by this performance work statement. The contractor will take every reasonable action to have the equipment repaired and operational within 24 hours after initial response by the contractor.

The contractor shall be responsible for furnishing maintenance, services, repairs and parts as required in order to maintain the system is operating in the proper working condition in accordance with the Original Equipment Manufacture's (OEM) specifications.

The contractor is required to maintain all equipment installed under this contract except for repairs necessitated by willful damage or negligence on the part of the U.S. Government.

The contractor shall provide full unscheduled and preventive maintenance service for the performance period at no additional cost to the Government. Service shall include repair and replacement of defective parts, complete maintenance program (as required by the manufacturer's manuals), and hotline telephone service to assist operators in troubleshooting problems.

Contractor must furnish all software upgrades issued by the equipment manufacturer.

Subject to mutual agreement between the Government and the Contractor, instrumentation may be upgraded within the specifications of the contract as new technology may become available. Additionally, the Government will add reagents through the life of this contract as a result of advances in technology or changes to clinical practice standards. Any such change to the reagents shall be mutually agreed upon by the contractor and the Government through a bilateral modification.

REPAIR

Upon notification of equipment failure, the Contractor shall respond to telephonic requests for unscheduled repair within one (1) working day. Should the analyzer become inoperable, the Contractor's technical specialist must be on-site within twenty-four (24) hours of initial notification. If deemed necessary after consultation with the contractor, if analyzer cannot be repaired within one (1) working day of the initial notification, contractor will work with GLWACH for determining the identification of alternative sources of testing. The Contractor will pay all costs associated with the provision of this service, for as long as the analyzer remains inoperable.

All work shall be performed in a professional manner by an authorized service representative or have two (2) years of maintenance experience on the contracted equipment. If any deficiencies are found due to

negligence of the service representative, the contractor shall be required to correct the deficiency to a fully operational status in accordance with manufacturer specifications at no additional cost to the U.S. Government.

Services are to be performed during the month(s) of October through September. Contractor shall respond within 48-hours, during normal duty hours of 8:00 AM and 4:00 PM excluding Federal Public Holidays.

Upon completion of services by the contractor's service representative, a written service report shall be provided to the laboratory point of contact. The service report shall provide detailed information regarding the cause of the equipment malfunction and corrective action taken. Include, at a minimum: 1) the time required to complete the work; 2) price of labor (hourly rate); 3) list of parts replaced with part numbers and prices for each part; 4) Nomenclature and Serial Number; 5) Date; and, 6) Name of the Technician. Service reports shall be emailed to the laboratory POC.

TRAINING

The Contractor will provide initial on-site operator training to all technicians assigned to the Pathology Laboratory prior to initiation of use of the analyzer for patient testing. In addition, the Contractor will provide at least two (2) training slots per analyzer at the Contractor's training site at no additional cost to the U.S. Government. The Contractor may charge the Government only for training services exceeding the basic requirement for a given year. Such training will be provided during designated implementation weeks and at mutually times, between the hours of 7 am - 7 pm local time.

Training shall include basic operation of the analyzers, troubleshooting procedures, performance of operator-level periodic preventive maintenance procedures and use of the data management/quality control software.